Kubernetes Cluster Deletion Guide

This document provides a safe, step-by-step procedure to completely delete a Kubernetes cluster created via **kubeadm**.

1. Drain and Delete Worker Nodes (From Control Plane)

Evict workloads safely and remove nodes from the cluster resource list before shutting them down:

```
# Safely evict all pods from the node
kubectl drain <node-name> --delete-emptydir-data --force --ignore-daemonsets

# Delete the node from the cluster
kubectl delete node <node-name>
```

2. Reset Worker Nodes

SSH into **each worker node** and revert the changes made by kubeadm join:

```
sudo kubeadm reset -f
```

3. Reset Control Plane Node(s)

SSH into your **control plane node(s)** and revert the cluster initialization changes:

```
sudo kubeadm reset -f
```

4. Manual Cleanup (Required on All Nodes)

The reset tool leaves behind network configuration, local databases, and credentials. Run these commands on **all machines**:

```
# Remove configuration files & directories
sudo rm -rf /etc/kubernetes/
sudo rm -rf /var/lib/etcd/
sudo rm -rf /var/lib/kubelet/
sudo rm -rf /etc/cni/net.d/

# Remove local kubectl configurations
rm -rf $HOME/.kube

# Reset network firewall rules (iptables)
sudo iptables -F && sudo iptables -t nat -F && sudo iptables -t mangle -F && sudo iptables
-X
```

5. Optional: Uninstall Kubernetes Packages

To completely remove the Kubernetes packages from your systems, use the respective package manager:

For Ubuntu / Debian systems:

```
sudo apt-get purge kubeadm kubectl kubelet kubernetes-cni kube* -y
sudo apt-get autoremove -y
```

For CentOS / RHEL systems:

```
sudo yum remove kubeadm kubectl kubelet kubernetes-cni kube* -y
sudo yum autoremove -y
```